

Worked examples for Lecture 12: Finite elasticity

David Al-Attar

Michaelmas Term, 2026

These examples accompany Lecture 12. They are intended as practice in the concrete calculations that underlie the lecture, and are less demanding than the questions on the problem sets. Numerical results were checked with the scripts in the python directory.

Example 1: Short descriptive title. Statement of the problem. Notation follows the lecture notes.

Solution. Solution text, with displayed equations as in the notes:

$$F_{ij} = \frac{\partial \varphi_i}{\partial x_j}. \quad (1)$$

Reference equations within this file as eq. (??). Results from the lecture are quoted by name (“the Christoffel equation”) rather than by number.